PRISM-R PRD-Lite

MediSched AI

PRISM Phase: R – Requirements Freeze
SmartX Technologies
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Purpose: Convert the approved problem into clear, testable, build-ready requirements. No solutioning or design decisions should be made in this document.

1. Background & Context

What problem are we solving?

Diagnostic centers rely on manual, phone-based workflows to schedule patient appointments and coordinate doctor availability. Patient information is scattered across paper files with no centralized registry. Doctors have no self-service mechanism to publish their consultation slots. Front-desk staff spend 2–4 hours/day on repetitive outbound calls for appointment confirmations and reminders. The result is high no-show rates (15–30%), frequent double-bookings, zero call documentation, and no operational analytics.

As documented in the approved PRISM-P Problem Brief, MediSched AI aims to eliminate these inefficiencies by providing: (1) a centralized patient registry managed by staff, (2) a doctor availability module where doctors self-manage consultation slots in 30-minute or 1-hour intervals, (3) a real-time calendar management system, (4) AI-powered outbound calling for patient engagement, (5) call transcription and documentation, and (6) operational analytics.

Why now?

- Manual scheduling is the #1 operational bottleneck reported by diagnostic center staff — it directly impacts revenue through no-shows and idle equipment.
- AI voice technology (Whisper, Twilio) has matured to the point where automated outbound calling is cost-effective and reliable for appointment-style interactions.
- The 8-week project timeline requires immediate requirements freeze to begin development within the 4-week prototype window.
- Competitive diagnostic centers are adopting digital scheduling — delay risks loss of operational advantage.

Reference: PRISM-P Problem Brief — MediSched AI (Approved, April 2026)

2. In-Scope

Features:

- Patient Management — centralized patient registry (create, read, update, search) managed by front-desk staff.
- Doctor Management — doctor profiles with specialization and status, managed by admin.
- Doctor Slot Management — doctors self-publish available consultation slots (30-min or 1-hour intervals) per day; admin fallback to manage slots on behalf of doctors.
- Calendar Management — real-time, multi-user appointment scheduling with daily/weekly/monthly views, conflict detection, and overbooking prevention.
- Automated Outbound Calling — AI/IVR-based patient calling with real-time slot fetching and booking during calls; three-attempt escalation with human fallback.
- Call Documentation — speech-to-text transcription of calls, auto-tagging notes to appointments, searchable and timestamped call logs.
- Automated Reminders — SMS and email reminders triggered by booked appointments.
- Analytics Dashboard — appointment utilization metrics, call success rates, no-show tracking, CSV export.
- User Roles & Access Control — Admin, Staff, Doctor, and Read-only roles with differentiated permissions.

Workflows:

- Patient registration and record management by staff.
- Doctor daily slot release and management.
- Appointment booking (manual by staff + automated via AI call).
- Outbound AI call flow with patient response capture and slot booking.
- Three-attempt call escalation with manual fallback.
- Automated reminder dispatch (SMS/email) post-booking.
- Call transcription, tagging, and log review.
- Admin analytics review and CSV data export.

User Roles:

- Admin, Staff (front-desk), Doctor, Read-only.

3. Out-of-Scope

- Integration with existing EHR/EMR or Hospital Information Systems (HIS).
- Multi-language voice AI support (prototype supports English only; architecture will be language-agnostic).
- Patient-facing mobile application (patients interact via voice calls and SMS only).
- Billing, invoicing, or payment processing for appointments.
- Insurance verification or pre-authorization workflows.
- Advanced AI diagnostics or clinical decision support.
- Multi-location / multi-branch diagnostic center support (single center for prototype).

- HIPAA/GDPR formal compliance certification (security best practices will be followed).
- Recurring weekly slot templates for doctors (manual daily slot release only in prototype).
- PDF export from analytics dashboard (CSV only in prototype).
- Patient self-registration (staff-managed only).

4. User Personas / Actors

4.1 Admin (Diagnostic Center Operations Manager)

- Oversees all scheduling operations, staff, and doctor management.
- Creates and manages doctor profiles (name, specialization, status).
- Can manage doctor slots as a fallback when doctors do not self-manage.
- Has full access to all patient records, call logs, and analytics.
- Exports operational reports (CSV).
- Manages user accounts and role assignments.

4.2 Staff (Front-Desk Personnel)

- Primary operator of day-to-day scheduling.
- Creates and maintains patient records (name, phone, DOB, gender, medical notes).
- Books appointments by selecting available doctor slots for patients.
- Reviews and edits call transcription notes tagged to appointments.
- Handles manual callback when AI call escalation occurs.
- Cannot manage doctor profiles or user accounts.

4.3 Doctor (Consultant / Specialist)

- Logs into the system to publish available consultation slots for the day or upcoming days.
- Selects date, time range, and slot duration (30 min or 1 hour) to release slots.
- Can block slots for breaks, emergencies, or unavailability.
- Views their own appointment schedule (booked patients for the day).
- Cannot access other doctors' schedules, patient management, or analytics.

4.4 Read-Only User

- Can view appointment schedules, patient records, and call logs.
- Cannot create, edit, or delete any data.
- Intended for supervisors or auditors who need visibility without modification rights.

4.5 System Actors

- AI Calling Engine — initiates outbound calls, captures patient responses, fetches available slots, and books appointments automatically.
- Reminder Service — sends automated SMS/email reminders based on booked appointment times.

- Transcription Service — converts call audio to text and tags transcription notes to the relevant appointment.

5. User Journeys / Use Cases

UC-01: Staff Registers a New Patient

Happy Path:

- Staff logs into MediSched AI and navigates to the Patient Management module.
- Staff clicks "Add New Patient" and enters required fields: full name, phone number, date of birth, gender.
- Staff optionally enters: email, address, medical notes, referring doctor.
- Staff clicks "Save". System validates required fields and creates the patient record.
- Patient record is now available for appointment booking and AI calling.

Alternative:

- If required fields are missing, the system displays inline validation errors and does not save.
- If a patient with the same phone number already exists, the system warns the staff and offers to view the existing record.

UC-02: Doctor Releases Available Consultation Slots

Happy Path:

- Doctor logs into MediSched AI and navigates to "My Availability".
- Doctor selects a date (today or future) and enters a time range (e.g., 9:00 AM to 12:00 PM).
- Doctor selects slot duration: 30 minutes or 1 hour.
- System auto-generates individual slots (e.g., 9:00–9:30, 9:30–10:00, ... for 30-min intervals).
- Doctor reviews the generated slots and confirms. Slots are now "Available" in the calendar.

Alternative:

- If any generated slot overlaps with an existing slot for the same doctor on the same date, the system flags the conflict and does not create overlapping slots.
- Doctor can block individual slots (e.g., for a break) by changing status to "Blocked".
- Admin can perform this workflow on behalf of a doctor via the Doctor Management module.

UC-03: Staff Books an Appointment for a Patient

Happy Path:

- Staff navigates to the Calendar module and selects a date.
- System displays available doctor slots (color-coded: green = available, blue = booked, grey = blocked).
- Staff selects an available slot and searches for a patient by name or phone number.
- System returns matching patient records. Staff selects the correct patient.
- Staff confirms the booking. Slot status changes from "Available" to "Booked" and links to the patient.
- System triggers automated SMS/email reminder to the patient based on the booked slot time.

Alternative:

- If the patient does not exist, staff is prompted to register a new patient (UC-01) before booking.
- If the slot was booked by another user in the meantime (concurrent access), the system displays a conflict error and refreshes available slots.

UC-04: AI Outbound Call to a Patient

Happy Path:

- System initiates an outbound AI call to a patient’s phone number from the Patients registry.
- An automated voice disclaimer plays: "This call may be recorded for quality and scheduling purposes."
- AI greets the patient and states the purpose (e.g., appointment reminder or new booking offer).
- Patient responds via voice prompts (e.g., "Press 1 to confirm, Press 2 to reschedule").
- If booking: AI fetches available doctor slots in real-time and offers time options to the patient.
- Patient selects a slot. AI confirms the booking. Slot status updates to "Booked".
- Call ends. Transcription service converts the call audio to text and tags it to the appointment.

Alternative — No Answer (Escalation):

- Attempt 1: AI calls the patient. No answer — system schedules Attempt 2 after 2 hours.
- Attempt 2: AI calls again. No answer — system escalates to manual staff callback.
- Attempt 3 (Manual): Staff receives a callback task and manually calls the patient.
- All attempts are logged with timestamps and outcomes for analytics.

Alternative — Patient Requests Human Agent:

- Patient says "speak to a person" or presses the fallback key during the AI call.

- System transfers the call to an available staff member (or queues a callback if none available).

UC-05: Admin Reviews Analytics Dashboard

Happy Path:

- Admin logs in and navigates to the Analytics module.
- Dashboard displays: appointment utilization rate, call success/failure rates, no-show count, per-doctor slot utilization.
- Admin filters by date range, doctor, or service type.
- Admin clicks "Export CSV" to download the filtered dataset.

Alternative:

- If no data exists for the selected filter range, the dashboard displays an empty state with a message: "No data available for the selected period."

UC-06: Staff Reviews and Edits Call Transcription

Happy Path:

- Staff navigates to the Call Logs module and searches by patient name, date, or appointment.
- System displays a list of call records with timestamp, patient name, call outcome, and transcription status.
- Staff selects a call record and views the full transcription text.
- Staff edits the transcription to correct any STT errors and adds manual notes.
- Staff saves. The updated transcription is linked to the appointment record.

6. Functional Requirements

Each requirement is assigned a unique ID for traceability through acceptance criteria, testing, and delivery.

6.1 Patient Management

ID	Requirement
FR-PM-01	The system shall allow Staff to create a new patient record with required fields: full name, phone number, date of birth, and gender.
FR-PM-02	The system shall allow Staff to optionally enter: email, address, medical notes, and referring doctor when creating or updating a

	patient record.
FR-PM-03	The system shall validate that full name and phone number are non-empty before saving a patient record.
FR-PM-04	The system shall warn Staff if a patient with the same phone number already exists and offer to view the existing record.
FR-PM-05	The system shall allow Staff to search patient records by name or phone number with partial match support.
FR-PM-06	The system shall allow Staff to update any field on an existing patient record.
FR-PM-07	The system shall display the appointment history and linked call transcription notes on the patient detail view.
FR-PM-08	The system shall restrict patient record creation and editing to Admin and Staff roles. Read-only users may only view.

6.2 Doctor Management

ID	Requirement
FR-DM-01	The system shall allow Admin to create a doctor profile with: full name, specialization, phone number, email, and active/inactive status.
FR-DM-02	The system shall allow Admin to update or deactivate a doctor profile. Inactive doctors shall not appear in the scheduling calendar.
FR-DM-03	The system shall allow Doctors to view and edit their own profile details (name, phone, email) but not their specialization or status.

6.3 Doctor Slot Management

ID	Requirement
FR-DS-01	The system shall allow a Doctor to release available consultation slots by selecting a date, start time, end time, and slot duration (30 minutes or 1 hour).
FR-DS-02	The system shall auto-generate individual slot entries based on the doctor's selected time range and duration (e.g., 9:00 AM–12:00 PM with 30-min duration = 6 slots).
FR-DS-03	Each slot shall have a status: Available, Booked, or Blocked.

FR-DS-04	The system shall prevent creation of overlapping slots for the same doctor on the same date.
FR-DS-05	The system shall allow a Doctor to block individual slots (setting status to Blocked) for breaks or emergencies.
FR-DS-06	The system shall allow Admin to create, modify, or block slots on behalf of any doctor.
FR-DS-07	The system shall update a slot's status from Available to Booked when a patient appointment is confirmed, and link the slot to the patient record.
FR-DS-08	The system shall prevent booking a slot that is already Booked or Blocked.

6.4 Calendar Management

ID	Requirement
FR-CM-01	The system shall display a calendar with daily, weekly, and monthly views showing all doctor slots with color-coded status indicators (Available = green, Booked = blue, Blocked = grey).
FR-CM-02	The system shall update the calendar in real-time when slots are created, booked, blocked, or cancelled by any user.
FR-CM-03	The system shall allow Staff to select an available slot on the calendar and book a patient into it.
FR-CM-04	The system shall detect and prevent double-booking: if two users attempt to book the same slot concurrently, only the first shall succeed and the second shall receive a conflict notification.
FR-CM-05	The system shall allow filtering the calendar view by doctor or service type (specialization).

6.5 Automated Outbound Calling

ID	Requirement
FR-AC-01	The system shall initiate outbound AI calls to patients using the phone number from their patient record.
FR-AC-02	The system shall play an automated voice disclaimer at the start of every AI call: "This

	call may be recorded for quality and scheduling purposes."
FR-AC-03	The system shall present voice prompts to the patient for appointment confirmation, rescheduling, or new booking.
FR-AC-04	The system shall fetch available doctor slots in real-time during the call and offer time options to the patient.
FR-AC-05	The system shall book the patient into the selected slot upon patient confirmation during the call and update the slot status to Booked.
FR-AC-06	The system shall implement a three-attempt escalation: Attempt 1 (AI), Attempt 2 (AI, after 2 hours), Attempt 3 (manual staff callback).
FR-AC-07	The system shall log every call attempt with: timestamp, patient ID, attempt number, call duration, outcome (answered/no-answer/booking-confirmed/escalated), and transcription reference.
FR-AC-08	The system shall transfer the call to an available staff member (or queue a callback) if the patient requests a human agent during the AI call.

6.6 Call Documentation & Transcription

ID	Requirement
FR-CD-01	The system shall transcribe every completed AI call using speech-to-text and store the transcription text.
FR-CD-02	The system shall auto-tag each transcription to the corresponding appointment and patient record.
FR-CD-03	The system shall allow Staff to view, search, and filter call logs by patient name, date, or appointment.
FR-CD-04	The system shall allow Staff to edit transcription text to correct speech-to-text errors.
FR-CD-05	The system shall allow Staff to add manual notes to any call log entry.
FR-CD-06	Each call log entry shall display: timestamp, patient name, call outcome, transcription text, and the user who last edited it.

6.7 Automated Reminders

ID	Requirement
FR-RM-01	The system shall send an automated SMS reminder to the patient’s phone number after an appointment is booked.
FR-RM-02	The system shall send an automated email reminder to the patient’s email address (if provided) after an appointment is booked.
FR-RM-03	The system shall send a follow-up reminder 24 hours before the appointment time.
FR-RM-04	The system shall log the delivery status of each reminder (sent, delivered, failed).

6.8 Analytics & Reporting

ID	Requirement
FR-AN-01	The system shall display an analytics dashboard with: appointment utilization rate (booked slots / total available slots), call success rate, no-show count, and per-doctor slot utilization.
FR-AN-02	The system shall allow Admin to filter analytics by date range, doctor, and service type.
FR-AN-03	The system shall allow Admin to export the filtered analytics data as a CSV file.
FR-AN-04	The system shall display an empty state message when no data matches the selected filters.

6.9 User Roles & Access Control

ID	Requirement
FR-UA-01	The system shall support four roles: Admin, Staff, Doctor, and Read-Only.
FR-UA-02	Admin shall have full access to all modules: patient management, doctor management, slot management, calendar, call logs, analytics, and user management.
FR-UA-03	Staff shall have access to: patient management (create/edit/view), calendar (book appointments), call logs (view/edit transcriptions). Staff shall not access doctor management, analytics export, or user

	management.
FR-UA-04	Doctor shall have access to: their own slot management (create/block slots), their own appointment schedule (view). Doctors shall not access patient records, other doctors' data, call logs, or analytics.
FR-UA-05	Read-Only users shall have view-only access to: calendar, patient records, and call logs. No create, edit, or delete permissions.
FR-UA-06	The system shall enforce role-based access on every request. Unauthorized actions shall return an access denied error.

7. Acceptance Criteria (Testable)

Acceptance criteria are written in Gherkin format (Given/When/Then) for each key functional requirement.

FR-PM-01: Create Patient Record

Given a Staff user is on the Patient Management page
When they fill in full name, phone number, date of birth, gender and click "Save"
Then a new patient record is created and appears in the patient list with all entered details.

FR-PM-04: Duplicate Phone Number Warning

Given a Staff user is creating a new patient with phone number "9876543210"
When a patient with the same phone number already exists in the system
Then the system displays a warning: "A patient with this phone number already exists" and offers a link to view the existing record.

FR-DS-01/02: Doctor Releases Slots

Given a Doctor is on the "My Availability" page
When they select date "2026-04-10", start time "09:00 AM", end time "12:00 PM", and duration "30 minutes" and click "Release Slots"
Then the system creates 6 individual slots (09:00, 09:30, 10:00, 10:30, 11:00, 11:30) each with status "Available" for that doctor on that date.

FR-DS-04: Overlapping Slot Prevention

Given a Doctor already has Available slots from 09:00 AM to 12:00 PM on 2026-04-10
When the Doctor attempts to release new slots from 11:00 AM to 01:00 PM on the same date
Then the system displays an error indicating overlap with existing slots (11:00 AM and 11:30 AM) and does not create the overlapping entries.

FR-CM-04: Concurrent Double-Booking Prevention

Given two Staff users are viewing the same available slot (Dr. Sharma, 10:00 AM, 2026-04-10)
When both users attempt to book the slot for different patients at the same time
Then only the first request succeeds (slot becomes "Booked"), and the second user receives a conflict notification: "This slot has already been booked. Please select another."

FR-AC-06: Three-Attempt Call Escalation

Given an AI outbound call to patient "Rahul Verma" goes unanswered (Attempt 1)
When 2 hours elapse
Then the system automatically initiates Attempt 2 (AI call). If Attempt 2 also goes unanswered, the system creates a manual callback task for Staff with patient details and call history.

FR-AC-05: Booking During AI Call

Given a patient is on an active AI call and selects "Book appointment" via voice prompt
When the AI presents available slots for the relevant doctor and the patient selects "10:30 AM, April 10th"
Then the slot is immediately booked (status changes to "Booked"), the patient receives a verbal confirmation, and the appointment is linked to the patient record.

FR-CD-01/02: Call Transcription and Auto-Tagging

Given an AI outbound call with patient "Priya Singh" has completed
When the transcription service processes the call audio
Then the transcription text is stored and automatically tagged to the appointment and patient record for "Priya Singh".

FR-RM-03: 24-Hour Reminder

Given a patient has a confirmed appointment at 10:00 AM on 2026-04-11
When the system clock reaches 10:00 AM on 2026-04-10 (24 hours before)
Then an SMS reminder is sent to the patient's phone number and an email reminder is sent to their email (if provided). Delivery status is logged.

FR-AN-03: CSV Export

Given an Admin is on the Analytics dashboard with filters applied (date range: April 1–10, doctor: Dr. Sharma)
When they click "Export CSV"
Then a CSV file is downloaded containing the filtered analytics data matching the dashboard display.

FR-UA-06: RBAC Enforcement

Given a Doctor user is logged in
When they attempt to access the Patient Management module
Then the system denies access and displays: "You do not have permission to access this module."

8. Edge Cases & Negative Scenarios

ID	Scenario	Expected Behavior
EC-01	Patient record with all optional fields empty	System creates the record with only required fields (name, phone, DOB, gender). Optional fields display as "Not provided" in the UI.
EC-02	Doctor releases slots for a past date	System rejects with error: "Cannot create slots for a past date."
EC-03	Doctor releases a single slot (start = 09:00, end = 09:30, duration = 30 min)	System creates exactly 1 slot. No error.
EC-04	Staff attempts to book a patient who has another appointment in an overlapping time window with a different doctor	System allows the booking (a patient may have appointments with multiple doctors). No conflict.

EC-05	AI call to an invalid/disconnected phone number	System logs the call attempt as "Failed — Invalid Number", does not retry this number, and creates a manual review task for Staff.
EC-06	STT transcription fails or returns empty text	System logs the call with transcription status "Failed". Staff can manually enter notes for the call record.
EC-07	Admin deactivates a doctor who has future booked slots	System warns Admin about existing future bookings. If Admin confirms deactivation, existing bookings remain valid but no new slots can be created for that doctor.
EC-08	Two doctors release slots at the exact same time range on the same date	System creates both sets of slots independently — no conflict (slots are per-doctor).
EC-09	SMS/email reminder delivery fails	System logs the failure status. Retry once after 15 minutes. If retry fails, log as "Reminder Failed" for admin visibility in analytics.
EC-10	Staff cancels an appointment after booking	Slot status reverts from "Booked" to "Available". Patient receives a cancellation notification (SMS). The cancelled booking is retained in history for analytics.
EC-11	Patient registry search returns zero results	System displays "No patients found" with an option to create a new patient record.
EC-12	Doctor attempts to block a slot that is already Booked	System rejects with error: "Cannot block a slot that has an active booking. Cancel the booking first."

9. Non-Functional Requirements (NFRs)

9.1 Performance

- **NFR-P-01:** Calendar page shall load within 2 seconds for up to 500 slots displayed.
- **NFR-P-02:** Patient search shall return results within 1 second for a database of up to 10,000 patient records.

- **NFR-P-03:** Slot booking operation shall complete within 1 second, including conflict detection and status update.
- **NFR-P-04:** Real-time calendar updates shall propagate to all connected users within 2 seconds of a slot status change.

9.2 Security

- **NFR-S-01:** All data shall be encrypted in transit (TLS 1.2+) and at rest (AES-256 or equivalent).
- **NFR-S-02:** Role-based access control shall be enforced on every API request. Unauthorized requests shall return HTTP 403.
- **NFR-S-03:** User sessions shall expire after 30 minutes of inactivity.
- **NFR-S-04:** All access to patient data and call recordings shall be audit-logged with user ID, timestamp, and action.
- **NFR-S-05:** Passwords shall be stored using a one-way hash with salt (bcrypt or equivalent).

9.3 Availability

- **NFR-A-01:** The system shall target 99% uptime during business hours (8 AM – 8 PM) for the prototype phase.
- **NFR-A-02:** Scheduled maintenance windows shall be communicated 24 hours in advance.

9.4 Scalability

- **NFR-SC-01:** The system shall support up to 20 concurrent users (staff + doctors) for the prototype phase.
- **NFR-SC-02:** The system shall support up to 10,000 patient records and 50,000 slot records without performance degradation.
- **NFR-SC-03:** The system shall handle up to 100 outbound AI calls per day for the prototype phase.

9.5 Observability

- **NFR-O-01:** The system shall log all API requests with: endpoint, user ID, timestamp, response status, and response time.
- **NFR-O-02:** The system shall log all call attempts with: patient ID, attempt number, timestamp, outcome, and duration.
- **NFR-O-03:** Error logs shall include stack traces and sufficient context for debugging.

9.6 Compliance

- **NFR-C-01:** Patient data shall only be accessible to authorized roles as defined in FR-UA-01 through FR-UA-06.
- **NFR-C-02:** Call recordings shall only be retained for the duration required by the diagnostic center’s data policy (default: 90 days). Configurable by Admin.
- **NFR-C-03:** An automated voice disclaimer shall precede every AI outbound call as defined in FR-AC-02.

10. Data & Integration Requirements

10.1 Data Entities

The following data entities are required to support the functional requirements. Detailed schema design will be addressed in PRISM-I (Implementation).

Entity	Key Information	Access
Patients	Full name, phone number, DOB, gender, email (opt), address (opt), medical notes (opt), referring doctor (opt), timestamps	Staff (create/edit), Admin (full), Read-only (view)
Doctors	Full name, specialization, phone, email, active/inactive status, timestamps	Admin (create/edit), Doctor (edit own profile)
Doctor Slots	Doctor reference, date, start time, end time, duration (30min/60min), status (Available/Booked/Blocked), patient reference (if booked), timestamps	Doctor (create/block own), Admin (full), System (update on booking)
Appointments	Patient reference, doctor slot reference, booking timestamp, booking source (manual/AI), status (confirmed/cancelled/no-show), reminder status	Staff (create/cancel), System (AI booking), Admin (full)
Call Logs	Patient reference, appointment reference, attempt number, timestamp, duration, outcome, transcription text, transcription status, last edited by	System (create), Staff (edit transcription), Admin (full)
Users	Name, email, role (Admin/Staff/Doctor/Read-only), hashed password, status, timestamps	Admin (create/manage)
Reminder Logs	Patient reference, appointment reference, channel (SMS/email), send timestamp, delivery status, retry count	System (create), Admin (view)

10.2 External Integrations

Integration	Purpose	Indicative Provider	Related FRs
Voice/Telephony Service	Outbound AI calls, IVR prompts, call recording	Twilio or equivalent Voice API	FR-AC-01 through FR-AC-08
Speech-to-Text Service	Transcription of call audio to text	OpenAI Whisper or Google STT	FR-CD-01
SMS Gateway	Appointment reminders and cancellation notifications	Twilio SMS or equivalent	FR-RM-01, FR-RM-03
Email Service	Email appointment reminders	SMTP provider or transactional email API	FR-RM-02, FR-RM-03

10.3 Data Relationships

- Each Appointment references one Patient and one Doctor Slot.
- Each Doctor Slot references one Doctor.
- Each Call Log references one Patient and one Appointment.
- Each Reminder Log references one Patient and one Appointment.
- A Patient can have many Appointments. A Doctor can have many Slots. A Slot can have at most one active booking.

11. Assumptions & Risks

11.1 Assumptions

- The PRISM-P Problem Brief decisions are final: PostgreSQL database, English-only prototype, simulated pilot environment, manual daily slot release.
- Staff will be the sole managers of patient records — patients do not self-register.
- Doctors will log in at least once daily to release their consultation slots. Admin fallback is available for non-compliant doctors.
- Mandatory patient fields (name, phone, DOB, gender) are sufficient for scheduling and calling. Optional fields do not block booking.
- Twilio free-tier supports the prototype’s estimated 50–100 daily outbound call volume.
- OpenAI Whisper or Google STT will achieve >85% transcription accuracy for English medical appointment conversations.
- The three-attempt call escalation (AI, AI, manual) is the agreed patient outreach strategy.
- CSV is the only export format required in the prototype. PDF export is deferred.
- The system will operate as a standalone platform with no EHR/EMR integration.

- Slot duration options are limited to 30 minutes and 1 hour. Custom durations are not supported.

11.2 Risks

ID	Risk	Likelihood	Impact	Mitigation
R-01	Low doctor adoption of daily slot release	Medium	High	Admin fallback (FR-DS-06); simple, quick slot release UI; daily reminder to doctors
R-02	AI call quality insufficient for patient trust	Medium	High	Human agent fallback (FR-AC-08); natural-sounding TTS; call quality testing before launch
R-03	STT transcription errors for medical terminology	Medium	Medium	Staff can edit transcriptions (FR-CD-04); iterative improvement of STT model
R-04	Concurrent booking conflicts under real-time load	Low	High	Conflict detection required (FR-CM-04); real-time calendar updates (FR-CM-02)
R-05	Twilio free-tier rate limits block testing	High	Medium	Test within limits; budget for paid tier if needed for demo/pilot
R-06	Scope creep beyond 8-week timeline	High	High	Strict adherence to in-scope/out-of-scope; weekly scope review
R-07	Patient data breach via insufficient access control	Low	Critical	RBAC from Day 1 (FR-UA-01–06); encryption (NFR-S-01); audit logging

				(NFR-S-04)

12. Open Questions

The following questions remain open at the requirements freeze stage and must be resolved during PRISM-I (Implementation) planning.

OQ-01: What is the exact SMS/email content template for appointment reminders and cancellation notifications?

Owner: Product Owner. Target resolution: Sprint 1 planning.

OQ-02: What is the maximum number of doctors the prototype needs to support simultaneously?

Owner: Engineering Lead. Estimate: 10–20 doctors for single-center prototype. To be confirmed during capacity planning.

OQ-03: Should the cancellation of an appointment trigger an AI call to the next patient on a waitlist?

Owner: Product Owner. Current decision: Deferred. Cancellation reverts slot to Available; no automatic waitlist in prototype.

OQ-04: What specific Twilio plan/tier will be used, and what are the exact rate limits for outbound calls?

Owner: Engineering Lead. To be confirmed during PRISM-I based on Twilio documentation and trial account setup.

OQ-05: What is the retention period for call recordings before automatic deletion?

Owner: Product Owner / Compliance. Default: 90 days (configurable by Admin). To be confirmed.

13. Definition of Done (DoD)

For this initiative to be considered complete, all of the following must be true:

- All functional requirements (FR-PM-01 through FR-UA-06) are implemented and pass their corresponding acceptance criteria.
- All edge cases (EC-01 through EC-12) are handled as specified.
- All non-functional requirements (Performance, Security, Availability, Scalability, Observability, Compliance) are met or explicitly deferred with documented rationale.

- Role-based access control is enforced across all modules for all four roles (Admin, Staff, Doctor, Read-only).
- Patient Management: Staff can create, search, update, and view patient records with validation and duplicate detection.
- Doctor Slot Management: Doctors can release 30-min and 1-hour slots, block slots, and view their schedule. Admin can manage slots on behalf of doctors.
- Calendar: Real-time, multi-user calendar displays slot statuses with conflict detection and color-coded views.
- Outbound Calling: AI calls execute with three-attempt escalation, human fallback, real-time slot booking, and voice disclaimer.
- Call Documentation: Calls are transcribed, auto-tagged to appointments, searchable, and editable by Staff.
- Reminders: SMS and email reminders are sent on booking and 24 hours before the appointment.
- Analytics: Dashboard displays utilization, call success, and no-show metrics with CSV export.
- All open questions (OQ-01 through OQ-05) are resolved or have documented deferral decisions.
- The system has been tested in the simulated environment with realistic data (mock patients, doctors, schedules).
- Product Owner has reviewed and signed off on the delivered functionality.

14. Approval & Freeze

Decision Owner Approval:

Name: ______________________________________

Signature / Approval: ______________________________________

Date: ______________________________________

By signing above, the Decision Owner confirms that:

- All in-scope requirements are reviewed and agreed.
- Out-of-scope items are acknowledged and will not be introduced without formal change request.
- Open questions have assigned owners and target resolution dates.
- This document serves as the requirements baseline for PRISM-I (Implementation) planning.

15. AI Gold Prompt Mapping (Mandatory)

This section defines the approved AI prompt structure to be used when drafting or refining this PRD-Lite. Free-form prompting is not allowed for PRISM-R artifacts.

15.1 Approved AI Tool(s)

- Primary: ChatGPT
- Secondary (feasibility validation only): Gemini Code Assist

15.2 Gold Prompt – PRISM-R

ROLE:

You are acting as a Product Owner and QA Lead at SmartX Technologies.

OBJECTIVE:

Convert the approved PRISM-P Problem Brief into a PRISM-R PRD-Lite with testable requirements.

CONTEXT:

- Company: SmartX Technologies
- Project type: Web Application (SaaS Platform)
- Scope boundaries: As defined in Sections 2 (In-Scope) and 3 (Out-of-Scope) of this document
- Known constraints: 8-week timeline, 4-week prototype, PostgreSQL, English-only, simulated pilot environment
- NFR expectations: Performance (2s page load), Security (encryption + RBAC), 99% uptime during business hours

INPUTS:

- Approved PRISM-P Problem Brief — MediSched AI (April 2026)
- User journeys: Patient registration, doctor slot release, appointment booking, AI outbound calling, call transcription review, analytics export
- Resolved decisions: PostgreSQL, 3-attempt call escalation, mandatory fields (name/phone/DOB/gender), admin slot fallback, CSV export only

INSTRUCTIONS:

- Do NOT propose solutions or architecture
- Write functional requirements with unique IDs
- Generate Gherkin-style acceptance criteria for each requirement
- Identify edge cases, assumptions, risks, and open questions

OUTPUT FORMAT:

- **1.** In-Scope / Out-of-Scope
- **2.** Functional Requirements
- **3.** Acceptance Criteria (Gherkin)
- **4.** Non-Functional Requirements
- **5.** Edge Cases
- **6.** Assumptions, Risks, Open Questions

QUALITY CHECK:

- Is every requirement testable?
- Are there any TBDs?
- Are NFRs explicitly stated or consciously deferred?

15.3 Evidence for PRISM Gate (M1 → M2)

- Completed PRISM-R PRD-Lite
- Acceptance criteria present for all functional requirements
- Decision owner sign-off
- Open questions tracked with owners